

Amazon RDS Read Replica Setup and Promotion

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon RDS** read replica was created from an existing primary database instance to support read scaling and backup-style failover use cases. The primary instance was selected from the RDS console, and a read replica was created in the same AWS Region with its own database identifier, instance class, security group, and Multi-AZ setting. After the replica was created, it received a separate endpoint from the primary instance, and all changes from the primary were asynchronously replicated to it. The lab also demonstrated how the read replica could be promoted to become an independent database instance. Finally, it was confirmed that after promotion, the new standalone database kept its own endpoint and no longer remained linked to the original primary instance.

Understanding Amazon RDS Read Replicas

An **Amazon RDS Read Replica** is a copy of a primary database that receives data changes asynchronously. It exists so applications can send read traffic to a separate database instance and reduce load on the primary database. This is helpful when reporting queries, analytics, or heavy read traffic would otherwise slow down the main database. A read replica can also be promoted later, which turns it into a standalone database instance.

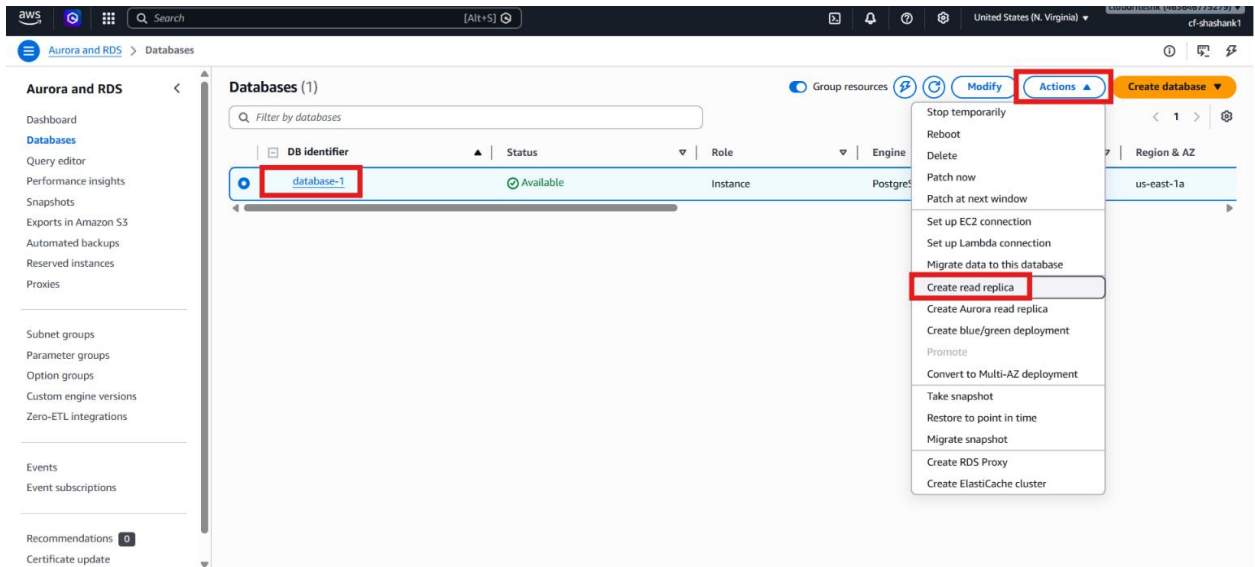
Example:

A production MySQL database handles both application writes and reporting queries. The application keeps writing to the primary database, while the reporting team connects to the read replica to run heavy SELECT queries. If the replica is promoted, it becomes its own database instance with a new endpoint.

Lab Steps

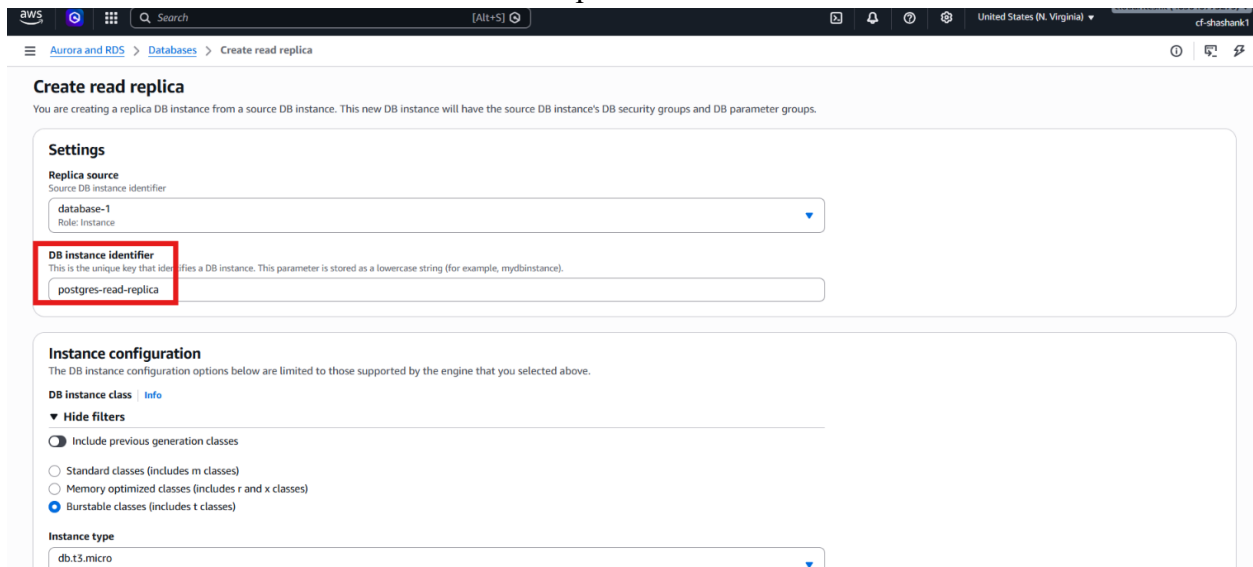
Step 1 – Select the Primary RDS Instance

- Sign in to the AWS Management Console.
- Open **Amazon RDS**.
- Create and select the primary database instance.
- Open **Actions**.
- Click **Create read replica**.



Step 2 – Configure the Read Replica

- Enter a new database identifier for the replica.



- Select the instance class for the replica.
- Choose any AWS Region but I will go for the same region for lab purpose.

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ **Hide filters**

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type
db.t3.micro

AWS Region

Destination Region
The Region where the replica will be launched.
US East (N. Virginia)

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

- Review the storage settings copied from the primary database.
- Keep storage auto scaling enabled if needed but I will disable.

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

For high-throughput workloads, we recommend provisioning at least 100 GiB of General Purpose (SSD) storage. Lower values might result in higher latencies when the initial I/O credit balance is used up. [Learn more](#)

Storage configuration upgrade [Info](#)

☐ Storage file system configuration upgrade
RDS recommends a storage file system configuration upgrade for your selected database instance.

You are on the latest storage configuration.

▼ **Additional storage configuration**

Storage autoscaling [Info](#)
Provides dynamic scaling support for your database's storage based on your application's needs.

☐ Enable storage autoscaling
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

- Enable **Multi-AZ** for the replica if high availability is required but for lab I will go with **Single-AZ**.

Availability

Deployment options [Info](#)

The following deployment options are limited to those supported by the engine.

☐ **Multi-AZ DB cluster deployment (3 instances)**
 Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
 • 99.95% uptime
 • Redundancy across Availability Zones
 • Increased read capacity
 • Reduced write latency

Write/read endpoint: AZ 1 (Primary instance + SSD)

Reader endpoints: AZ 2 (Readable standby + SSD), AZ 3 (Readable standby + SSD)

☐ **Multi-AZ DB instance deployment (2 instances)**
 Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
 • 99.95% uptime
 • Redundancy across Availability Zones

Write/read endpoint: AZ 1 (Primary instance)

Standby (no endpoint): AZ 2 (Standby)

☒ **Single-AZ DB instance deployment (1 instance)**
 Creates a single-AZ DB instance with no reader instances. This setup provides:
 • 99.5% uptime
 • No data redundancy

Write/read endpoint: AZ 1 (Primary instance)

- Select the appropriate DB subnet group.
- Use the same security group as the primary database.

Connectivity

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ **Dual-stack mode**
Your resources can communicate over IPv4, IPv6, or both.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-099792461c7d0a616

Public access

☒ **Publicly accessible**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **Not publicly accessible**
No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Existing VPC security groups

Choose VPC security groups

default X RDS-SG X

Availability Zone [Info](#)

The EC2 Availability Zone that the database will be created in.

No preference

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

- Keep the same authentication settings.
- Create the read replica.

Copy tags to snapshots

☒ Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console.

AWS KMS key [Info](#)
Enter a key ARN

Amazon Resource Name (ARN)
arn:aws:kms:us-east-1:463646775279:key/fd9c9c8d-03ce-425a-917f-1d2e959cf13e
Example: arn:aws:kms:<region>:<accountID>:key/<key-id>

Account
463646775279

KMS key ID
fd9c9c8d-03ce-425a-917f-1d2e959cf13e

Maintenance
Auto minor version upgrade [Info](#)
☒ Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).
☐ Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

[Cancel](#) [Create read replica](#)

Step 3 – Wait for the Replica to Become Available

- Wait while RDS creates a snapshot of the primary database.
- Allow RDS to create the replica from the snapshot.
- Wait for the replica status to change to **Available**.

Successfully created replica *postgres-read-replica* in US East (N. Virginia)

Databases (2)

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout ...	Region & AZ	Size	R
database-1	Available	Primary	PostgreSQL	SECOND	us-east-1a	db.t3.micro	
postgres-read-replica	Available	Replica	PostgreSQL	SECOND	us-east-1b	db.t3.micro	

- Confirm that the replica has a different endpoint from the primary database.

Step 4 – Verify Read Replica Connectivity

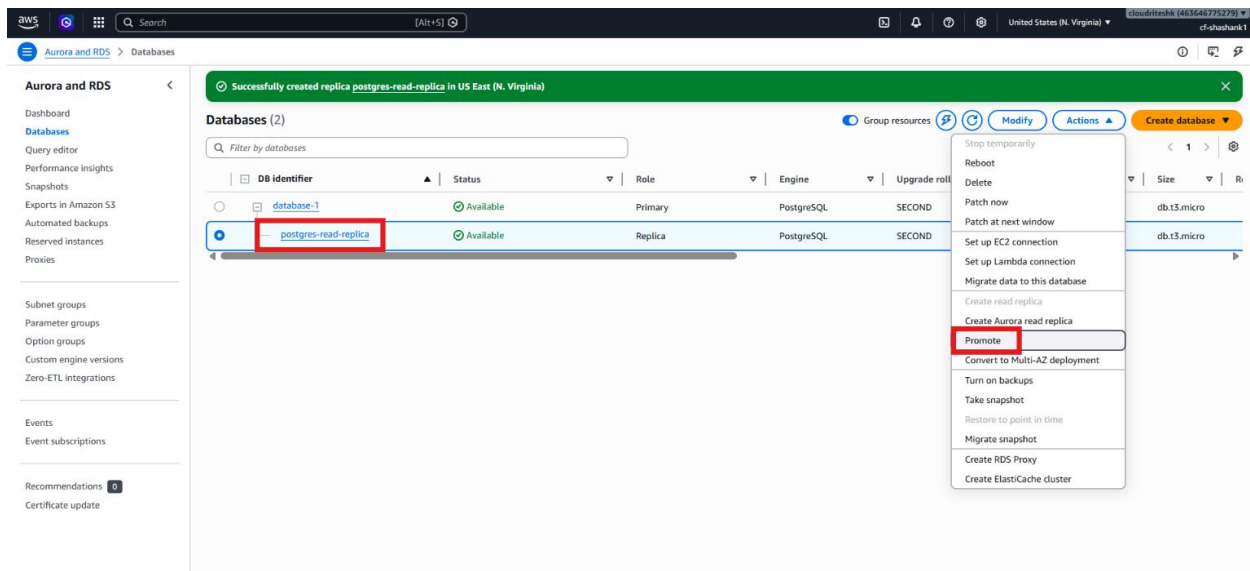
- Open the primary database endpoint.

- Open the read replica endpoint.

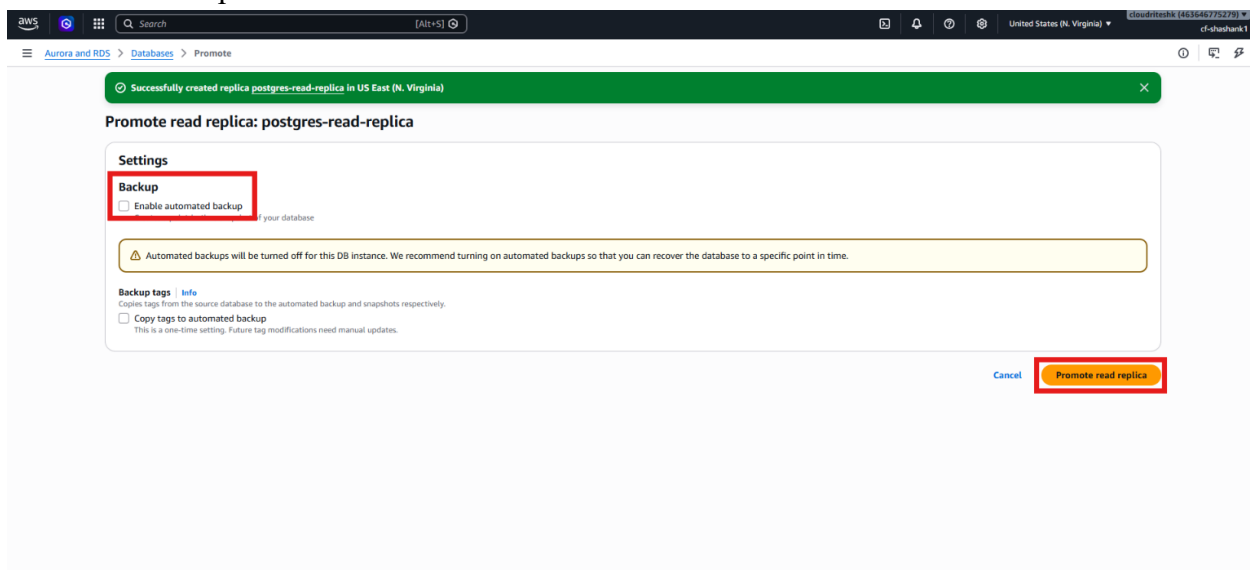
- Verify that the replica has its own DNS name.

Step 5 – Promote the Read Replica

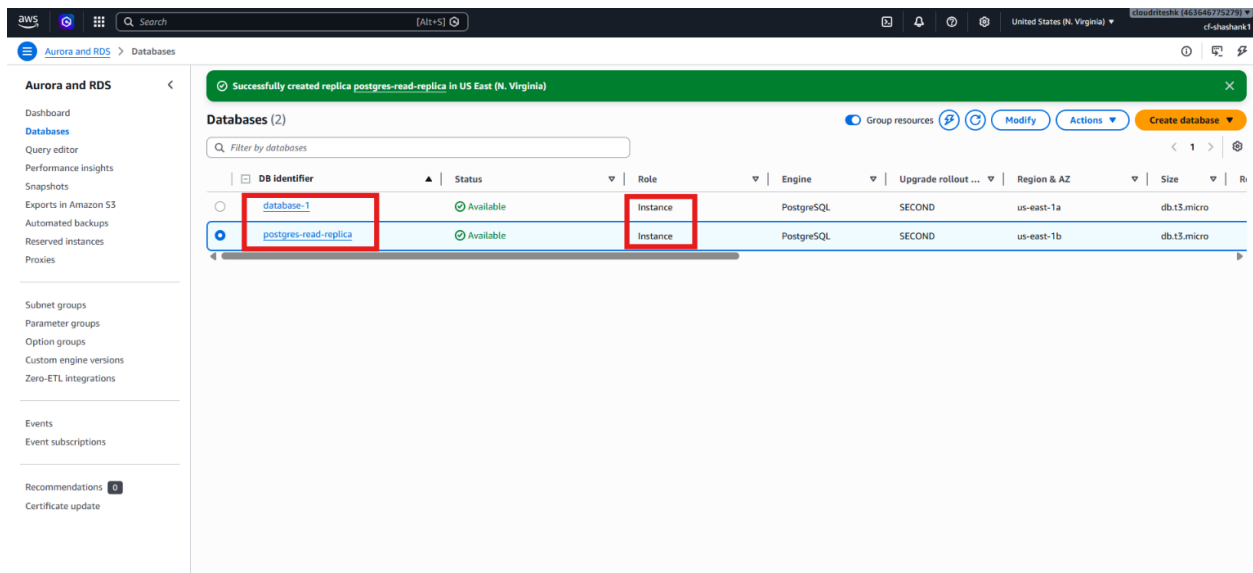
- Open the read replica instance in **Amazon RDS**.
- Click **Actions**.
- Choose **Promote read replica**.



- Enable automated backups if required but I will not select it.
- Confirm the promotion.

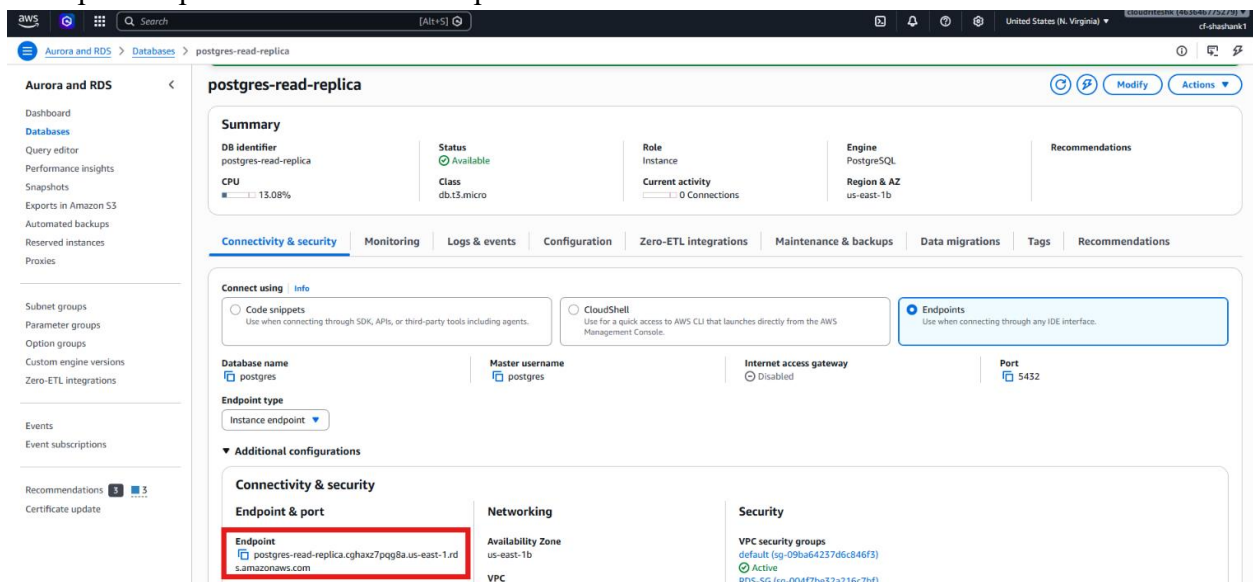


- Wait for the replica to reboot and finish promotion.
- Verify that the promoted instance is now independent and no longer linked to the primary database.



Step 6 – Verify the Promoted Database

- Open the promoted database endpoint.



- Confirm that it has a different endpoint from the original primary database.

The screenshot displays the AWS Management Console for an Amazon RDS instance named 'database-1'. The instance is a PostgreSQL db.t3.micro in the us-east-1a region. The 'Connectivity & security' tab is active, showing the 'Endpoint & port' section with the endpoint 'database-1.cghaxz7pqg8a.us-east-1.rds.amazonaws.com' highlighted in a red box. Other sections include 'Networking' (Availability Zone: us-east-1a) and 'Security' (VPC security groups: default (sg-09ba64237d6c846f3)).

- Verify that it now behaves as a standalone RDS instance.
- Reconfigure any application or client connections to point to the new endpoint.

Verification

- The read replica was created successfully from the primary **Amazon RDS** instance.
- The replica received its own endpoint.
- Data changes from the primary instance replicated asynchronously to the replica.
- The read replica was promoted successfully.
- The promoted replica became an independent database instance.
- The new standalone database used a different endpoint and required client reconfiguration.